VS-01 Data Acquisition and Ingestion

Table of contents

VS-01 Data Acquisition and Ingestion . . . 1
1. Purpose and value created . . . 1
2. Position and state transition . . . 1
3. Stages . . . 2
4. The four process types . . . 6
5. Actors and roles . . . 15
6. Capabilities, processes, and functions . . . 16
7. Principles . . . 18
8. Interfaces, services, and events . . . 20
9. Key performance indicators . . . 21
10. Examples . . . 22

VS-01 Data Acquisition and Ingestion

1. Purpose and value created

VS-01 is the entry point of the statistical value chain. It brings data from outside the NSI under NSI custody: survey responses, administrative registers, exchanges in data spaces, and new sources such as web or sensor data. It does not change the content of the data. It makes sure the data is obtained lawfully, received securely, checked for integrity, and registered with its provenance. The value created is **trust and readiness**: trust that the data was acquired correctly, and readiness for use in the rest of the chain. Everything downstream depends on this. If the acquisition of a source cannot be traced, no statistic built on it can be fully defended.

VS-01 corresponds to the business domain **Observation/Collection** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

PLACEHOLDER: ArchiMate view VS-01 (a) value stream and capabilities

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2. Position and state transition

VS-01 is the first primary value stream. All data from outside the NSI enters the statistical system through it, before any downstream processing takes place.

VS-01 takes data from boundary position #0 to State #1. Before VS-01, the data is held by a provider or respondent, outside the NSI's control. After VS-01, it is a registered, immutable version of raw data, in NSI custody, in its original form, with provenance metadata recorded and basic quality verification completed. The quality gate, the minimum metadata set, and the protection of State #1 are described in *State #1: Raw Data* and are not repeated here. **[VC10]** VS-01 is the only entrance for data from outside the NSI: there is no side door into the chain. A product that starts from data already registered by another domain does not pass VS-01; it enters the chain downstream.

From	To	Gate	Event out
#0 Source Data	#1 Raw Data	Gate of State #1 (legal basis, authenticity, integrity, security, structure, completeness, timeliness, metadata)	*Raw Data Registered*

PLACEHOLDER: ArchiMate view VS-01 (c) state handover and events: Source Data (#0) outside the NSI boundary, the ingestion service at the boundary (metadata addition, quality and schema check), Raw Data (#1) inside the NSI

3. Stages

The stages describe the value that VS-01 adds in every production cycle. Together they form the execution process type of VS-01. In a mature steady state, most of the work has been designed and built in advance (see attribute 4). Execution then focuses on running the cycle, capturing data and metadata, and making sure the quality gate is passed.

Stage	Name	Value added
VS-01.1	Identify Data Needs and Sources	The information need is clear, and suitable sources are known
VS-01.2	Establish Data Agreements and Access Rights	Data can be obtained lawfully, under agreed conditions
VS-01.3	Collect or Receive Data	Data and its metadata are in NSI custody
VS-01.4	Validate and Log Data Intake	The delivery is complete, intact, and traceable
VS-01.5	Store and Register Raw Data Artefacts	Raw data is registered and available for reuse (State #1)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies. **[VC9]**

VS-01.1 Identify Data Needs and Sources

Value created: *Clear understanding of what data is needed and where it can be obtained.*

This stage starts from the information need set in the design of the statistical product. In a mature steady state, the Design process *Design Data Requirements and Source Strategy* has already defined the data requirements and the source strategy. Execution in each cycle therefore focuses on confirmation rather than re-creation. The team confirms or updates the data requirements for the current production cycle. It verifies that the identified sources remain available and accessible, and checks that source metadata is current: contact persons, data formats, and delivery schedules. Before any new collection, it checks whether the data already exists as a registered state version, in line with SAF-P08. It also identifies new sources that may have become available since the last cycle. A significant change in requirements or source availability goes back to design instead of being solved during the cycle.

Activity	Key outputs	SAF process	Other reference
Confirm or update data requirements for this production cycle	Confirmed requirements for the cycle	PRD-1.1 Identify needs	

Activity	Key outputs	SAF process	Other reference
Check whether the data already exists as a registered state version	Reuse decision	PRD-1.5 Check data availability and suitability	SAF-P08
Verify data source availability and accessibility	Source availability confirmation	PRD-1.5 Check data availability and suitability	
Check that source metadata is current (contacts, formats, schedules)	Updated source metadata	PRD-1.5 Check data availability and suitability; OVA-6 Data Supplier Management	
Identify new sources available since the last cycle	New source assessment (if any)	PRD-1.5 Check data availability and suitability	

VS-01.2 Establish Data Agreements and Access Rights

Value created: *Legal, ethical, and transparent access to data under controlled conditions.*

Before data flows, legal and operational access must be confirmed for the current production cycle. For each source, the legal basis is established: a statutory obligation, a data provision agreement, or a usage policy in a data space. The team activates or renews data sharing agreements. It verifies compliance with data protection law for the current data scope, for example by checking that the records of processing activities (GDPR Article 30) are up to date. It confirms that access credentials and security certificates are valid and have not expired. For sources that are new in this cycle, the full agreement process is carried out. The active agreement status is recorded and linked to the upcoming deliveries. This creates a traceable chain from legal authorisation to received data, so the legal basis of each version can be shown in an audit.

Activity	Key outputs	SAF process	Other reference
Activate or renew data sharing agreements for the current cycle	Active agreement record (per source)	OVA-6 Data Supplier Management; DSP-3 Agree data use conditions	3.4.3 Service agreement
Verify data protection compliance: records of processing up to date	Compliance confirmation log	DLM-4 Classify and protect data	SAF-P11
Confirm that access credentials and security certificates are valid	Verified access credentials	PRD-4.2 Set up collection	SAF-P10
For new sources: carry out the full agreement process	New source agreement	OVA-6 Data Supplier Management; DSP-1 Join a data space; DSP-3 Agree data use conditions	SAF-P11
Record the active agreement status, linked to upcoming deliveries	Agreement-delivery link	PRD-4.2 Set up collection	

VS-01.3 Collect or Receive Data

Value created: *Secure and verifiable transfer of authentic data into the organisation.*

This stage is the operational heart of the value stream, where data physically enters NSI custody. Primary data collection runs surveys through the designed and built instruments. Web questionnaires (CAWI) open, invitation links are sent, and response rates are monitored. Field interviewers (CAPI) are deployed with managed assignments, and telephone interview schedules (CATI) are carried out with callback management. Non-response is followed up within the agreed strategy. Secondary data collection receives administrative

data through agreed channels. Secure file transfer (SFTP) accepts scheduled deliveries, APIs pull data on agreed schedules, and data space connectors support federated exchange. Emerging sources, such as web scraping, sensor data, and commercial feeds, are extracted according to their specification.

Equally critical is the metadata captured at the moment of receipt. For every dataset received, the ingestion process records the fields in the table below. This obligation follows from SAF-P09. It makes sure that no data enters the NSI without the context needed to understand its provenance and quality. At the end of this stage, the data is held by the NSI, but it has not yet been accepted.

Activity	Key outputs	SAF process	Other reference
Create the frame and select the sample for this cycle (surveys)	Frame and sample	PRD-4.1 Create frame and select sample	1.2.1 Primary data collection
Prepare instruments, field staff, and channels; plan the collection period	Collection ready to start	PRD-4.2 Set up collection	
Run survey collection and follow up non-response	Survey responses, paradata	PRD-4.3 Run collection	1.2.1 Primary data collection
Receive administrative data and data space deliveries through agreed channels	Received deliveries	PRD-4.3 Run collection; DSP-4 Exchange data under usage control	1.2.2 Secondary data collection
Extract data from emerging sources according to specification	Extracted data	PRD-4.3 Run collection	
Capture ingestion metadata and paradata at the point of receipt	Ingestion metadata record	PRD-4.3 Run collection; OVA-4 Process Data Management	SAF-P09

Ingestion metadata field	Description
Source identifier and organisation	Who provided the data
Timestamp of receipt	When it was received
Transfer method and channel	How it was delivered (SFTP, API, CAWI, and so on)
File hash (SHA-256) and format	Integrity verification and format record
Volume (record count, file size)	Scale of the delivery
Agreement reference	Legal basis for the data
Collection mode and instrument version	Survey-specific provenance

PLACEHOLDER: VS-01 collection channel architecture: survey, administrative, and emerging-source channels arriving at one ingestion point, each adding its own metadata (source, timestamp, transfer method, file hash, volume, agreement reference), followed by VS-01.4

VS-01.4 Validate and Log Data Intake

Value created: *Confidence that received data is complete, structured, and fully traceable.*

Once data has been received, it must pass the quality gate of State #1 before it can be accepted. The validation checks completeness: expected record counts are present, required fields are populated, and the delivery matches the service level agreement. It checks structure and format against the raw data schema from *Design Data Model and Metadata Schema.* It checks security and integrity: virus scan, checksum

verification (SHA-256), and digital signature validation, in line with SAF-P10. A quality assessment is carried out against the criteria from *Design Validation Rules and Quality Criteria*, producing a completeness rate, a timeliness score, and a conformance rating.

Every validation result, exception, and decision is logged in an audit trail. The trail captures data metadata (what was validated, what passed, what failed) and process metadata (who validated, when, which rules were applied, which tool version). The outcome is one of two business events. *Data Intake Validated* starts storage and registration. *Data Intake Rejected* starts exception handling according to the designed rules: notify the provider through the relationship manager, queue the delivery for manual review, or accept it conditionally with a quality flag.

No content is edited in this stage: only technical and structural validation is done. Content editing, such as correction, imputation, and harmonisation, belongs to VS-03 Data Editing and Enrichment.

Activity	Key outputs	SAF process	Other reference
Verify completeness: record counts, required fields, conformance with the service level agreement	Completeness assessment	PRD-4.4 Finalise collection	
Validate structure and format against the raw data schema	Structural validation result	PRD-4.4 Finalise collection; DLM-2 Promote data through a quality gate	
Perform security and integrity checks: virus scan, checksum, signature	Security verification result	PRD-4.4 Finalise collection	SAF-P10
Carry out the quality assessment against the designed criteria	Quality assessment scores	DLM-2 Promote data through a quality gate	
Create the audit trail: log all validation results, exceptions, and decisions	Audit trail record	OVA-4 Process Data Management	GAMSO Quality Management

VS-01.5 Store and Register Raw Data Artefacts

Value created: *Secure, organised, and traceable storage of raw data.*

This stage persists validated data and makes it findable for the rest of the statistical system. Each dataset version receives a unique persistent identifier (URI or UUID). It is stored in the raw data repository in its original format, in line with the loose coupling of SAF-P07. Its metadata is registered in the data catalogue, in line with SAF-P09. Each dataset is linked to its provenance record (source, method, timing), its quality assessment from VS-01.4, the governing data sharing agreement, and the classification and coding schemes it uses. Lineage from Source Data (#0) to Raw Data (#1) is recorded. Access permissions are applied according to the access control and privacy design and the decisions of the data owner. A redelivery or correction by the provider always creates a new version; it never overwrites an earlier one. The stage ends by publishing the *Raw Data Registered* event, which tells VS-02 that new raw data is available for standardisation.

Output: Raw Data (State #1), the official handover artefact to the rest of the statistical value chain. Its structure and minimum metadata set are described in *State #1: Raw Data*. **[VC10]**

Activity	Key outputs	SAF process	Other reference
Assign a unique persistent identifier (URI or UUID) to the dataset version	Dataset ID	DLM-3 Register and catalogue data	

Activity	Key outputs	SAF process	Other reference
Store the data in the raw data repository in its original format	Stored raw data artefact	DLM-5 Store and version data	SAF-P07
Register metadata in the data catalogue: provenance, quality, agreement, classification, lineage	Catalogue registration	DLM-3 Register and catalogue data; OVA-2 Metadata Management	SAF-P09
Apply access permissions according to the designed framework	Access-controlled dataset	DLM-4 Classify and protect data	SAF-P10, SAF-P11
Publish the *Raw Data Registered* event to notify VS-02	Event notification	DLM-3 Register and catalogue data	SAF-P07

4. The four process types

The sub-processes below are specific to VS-01. Each refines one or more SAF business processes, listed as *Refines*, which keeps it traceable to *Business Processes*. Sub-processes marked *(standard)* work as described in Standard activities in every value stream; only the VS-01 specifics are listed.

Design: provides metadata

Design sub-process	Refines	Main outputs
Design Data Requirements and Source Strategy	PRD-1.1 Identify needs; PRD-1.2 Consult and confirm needs; PRD-1.3 Establish output objectives; PRD-1.5 Check data availability and suitability; PRD-1.6 Prepare and submit business case; PRD-2.3 Design collection	Data requirements specification, source assessment, collection strategy
Design Collection Methodology and Instruments	PRD-2.2 Design variables; PRD-2.3 Design collection; PRD-2.4 Design frame and sample	Collection methodology, questionnaire design, sample design, communication plan
Design Data Model and Metadata Schema	PRD-2.2 Design variables; DLM-1 Plan and design data assets; OVA-2 Metadata Management	Raw data schema, ingestion metadata model, catalogue template
Design Validation Rules and Quality Criteria	PRD-2.5 Design processing and analysis; DLM-2 Promote data through a quality gate; OVA-1 Quality Management	Validation rule set, quality criteria, exception handling policy
Design Access Control and Privacy Framework	DLM-1 Plan and design data assets; DLM-4 Classify and protect data	Data classification, access control design, privacy impact assessment

Design Data Requirements and Source Strategy

Refines: PRD-1.1 Identify needs; PRD-1.2 Consult and confirm needs; PRD-1.3 Establish output objectives; PRD-1.5 Check data availability and suitability; PRD-1.6 Prepare and submit business case; PRD-2.3 Design collection.

The starting point of any data acquisition effort is understanding what data is needed and where it can be obtained. This sub-process translates statistical objectives into a concrete source strategy: target

populations, required variables, reference periods, and quality expectations. The team inventories and evaluates potential sources across the full spectrum: surveys, administrative registers, APIs, web sources, sensors, data spaces, and commercial data. Each source is assessed for relevance, quality, accessibility, cost, timeliness, and legal availability. Reuse of existing data comes before new collection, in line with SAF-P08. The outcome is a decision on the collection approach: primary collection (run a survey), secondary collection (reuse administrative data), or a combination of sources. The trade-offs behind that decision are documented.

Activity	Key outputs	SAF process	Other reference
Define statistical needs: target population, variables, reference period, quality expectations	Data requirements specification	PRD-1.1 Identify needs; PRD-1.2 Consult and confirm needs; PRD-1.3 Establish output objectives	1.1.1 Needs inventory
Inventory and evaluate potential data sources of all types	Source assessment report	PRD-1.5 Check data availability and suitability	SAF-P08
Assess each source for relevance, quality, accessibility, cost, timeliness, legal availability	Source assessment report	PRD-1.5 Check data availability and suitability	
Decide the collection strategy: primary, secondary, or multi-source	Collection strategy decision record	PRD-2.3 Design collection	
Document the trade-offs between source options	Collection strategy decision record; business case	PRD-1.6 Prepare and submit business case	

Design Collection Methodology and Instruments

Refines: PRD-2.2 Design variables; PRD-2.3 Design collection; PRD-2.4 Design frame and sample.

Once the source strategy is set, the methodology team designs how data will actually be collected. This involves selecting collection modes and defining how each mode works within the overall design. Examples are web, face-to-face, or telephone interviewing (CAWI, CAPI, CATI), secure file transfer (SFTP), automated machine-to-machine interfaces (API), data space connectors, and web scraping. For survey-based collection, the design includes the questionnaire: variable definitions, question wording, skip logic, and code lists. The sampling frame and sample selection method are designed to make the sample statistically representative. Where several modes are used, the team specifies which respondents receive which mode and the fallback sequence for non-response. A respondent communication strategy completes the design: invitation letters, reminders, and help desk scripts. Respondent burden is kept as low as possible.

Activity	Key outputs	SAF process	Other reference
Design collection modes (CAWI, CAPI, CATI, SFTP, API, data space connector, web scraping)	Collection methodology document	PRD-2.3 Design collection	1.2.1 Primary data collection; 1.2.2 Secondary data collection
Design questionnaire content: variable definitions, question wording, skip logic, code lists	Questionnaire design (per mode)	PRD-2.2 Design variables; PRD-2.3 Design collection	
Design the sampling frame and sample selection method	Sample design and selection criteria	PRD-2.4 Design frame and sample	

Activity	Key outputs	SAF process	Other reference
Define the multi-mode strategy and fallback sequences	Collection methodology document	PRD-2.3 Design collection	
Design the respondent communication strategy	Respondent communication plan	PRD-2.3 Design collection	3.4.2 Contact management

Design Data Model and Metadata Schema

Refines: PRD-2.2 Design variables; DLM-1 Plan and design data assets; DLM-9 Manage reference and master data; OVA-2 Metadata Management.

This sub-process defines the target data structure of Raw Data (State #1) and the metadata that must accompany every dataset at the point of ingestion. The data model specifies fields, types, formats, and constraints for the storage area where incoming data first lands, the landing zone. Equally important is the ingestion metadata model. It prescribes what must be captured at each point of receipt: provenance (source organisation, contact, agreement reference), method (collection mode, instrument version, sampling parameters), timing (collection period, delivery and processing timestamps), and quality indicators (completeness rate, delivery timeliness, format conformance). Classification and coding schemes are defined, consistent with national and international standards. Finally, the structure of the metadata catalogue is designed so that datasets are findable, linked, and traceable. This makes active metadata a foundation for automation downstream, as emphasised by the Metadata Management capability of CSDA (Common Statistical Data Architecture).

Activity	Key outputs	SAF process	Other reference
Define the target data structure of Raw Data (State #1): fields, types, formats, constraints	Raw data schema definition	PRD-2.2 Design variables; DLM-1 Plan and design data assets	
Define the ingestion metadata model: provenance, method, timing, quality indicators	Ingestion metadata model	OVA-2 Metadata Management	SAF-P09
Define classification and coding schemes	Classification mapping rules	DLM-9 Manage reference and master data	SAF-P09
Design the metadata catalogue structure for registration and discovery	Catalogue registration template	DLM-1 Plan and design data assets; OVA-2 Metadata Management	CSDA Metadata Management

Design Validation Rules and Quality Criteria

Refines: PRD-2.5 Design processing and analysis; DLM-2 Promote data through a quality gate; OVA-1 Quality Management; OVA-6 Data Supplier Management.

Before any data enters the NSI, the acceptance criteria and the rules for handling exceptions must be clear. This sub-process sets completeness thresholds, such as minimum record counts and required field coverage. It sets the parameters of service level agreements: delivery windows, response times, and format specifications. It defines structural validation rules: field types, value ranges, referential integrity, and format conformance. Security and integrity checks are specified in line with SAF-P10: virus scan requirements, checksum algorithms, and encryption standards. The quality dimensions to track are agreed: completeness, timeliness, conformance, and integrity. Exception handling rules distinguish between rejection, conditional acceptance with quality flags, and routing to manual review. Together, these rules make up the VS-01 part of the quality gate of State #1. The Quality Management capability of GAMSO provides the overall control framework.

Activity	Key outputs	SAF process	Other reference
Define completeness thresholds and service level parameters	Quality criteria document, service level template	PRD-2.5 Design processing and analysis; OVA-6 Data Supplier Management	3.4.3 Service agreement
Define structural validation rules: field types, value ranges, referential integrity	Validation rule set	PRD-2.5 Design processing and analysis	
Define security and integrity check specifications	Validation rule set	PRD-2.5 Design processing and analysis	SAF-P10
Define the quality dimensions to track	Quality criteria document	OVA-1 Quality Management	GAMSO Quality Management
Design exception handling rules: rejection, conditional acceptance, manual review	Exception handling policy	DLM-2 Promote data through a quality gate	

Design Access Control and Privacy Framework

Refines: DLM-1 Plan and design data assets; DLM-4 Classify and protect data; OVA-4 Process Data Management.

This last design sub-process makes sure that security and privacy are built in from the start, not added afterwards. Each data source receives a classification for availability, integrity, and confidentiality, in line with SAF-P09. Role-based access controls for the raw data repository are designed in line with SAF-P10: who can read, write, and administer. Privacy protection follows SAF-P11. It covers the moment of pseudonymisation (when direct identifiers are removed or replaced), purpose limitation (which uses each agreement allows), the legal basis for processing (for example, which ground of GDPR Article 6 applies to each source), and data minimisation (only necessary data is collected). Audit logging requirements, which define the access events that must be recorded, complete the design.

Activity	Key outputs	SAF process	Other reference
Define the data classification (availability, integrity, confidentiality) per source	Data classification policy	DLM-4 Classify and protect data	SAF-P09
Design role-based access control for the raw data repository	Access control design	DLM-1 Plan and design data assets; DLM-4 Classify and protect data	SAF-P10; 3.3.4 Logical identity and access control
Design privacy protection: moment of pseudonymisation, purpose limitation, legal basis, data minimisation	Privacy impact assessment	DLM-4 Classify and protect data	SAF-P11; 3.3.1 Privacy management
Design audit logging requirements	Audit logging specification	OVA-4 Process Data Management	SAF-P10

PLACEHOLDER: VS-01 design process decomposition: inputs (statistical needs, data sources, framework standards, legal and privacy requirements), the five design sub-processes in order, and the design document each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

Implementation sub-process	Refines	Main outputs
Build Collection Instruments and Channels	PRD-3.1 Reuse or build collection instruments; DSP-1 Join a data space	Production-ready instruments and channels
Configure Data Ingestion Workflows	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Automated validation, metadata capture, exception routing, events
Build Metadata Registration and Cataloguing	PRD-3.4 Configure workflows; DLM-3 Register and catalogue data; OVA-2 Metadata Management	Catalogue, automated metadata propagation, discovery
Test and Validate the Pipeline *(standard)*	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalise production systems	Test reports; field test with actual data providers; penetration test of transfer channels; volumes tested against the agreed delivery windows

Build Collection Instruments and Channels

Refines: PRD-3.1 Reuse or build collection instruments; DSP-1 Join a data space.

This sub-process turns the collection methodology into operational tools and channels. For survey-based collection, it builds or configures the instruments. A CAWI portal gets the designed question logic, support for several languages, and a layout that works on any device. A CAPI application gets offline capability and the designed questionnaire. CATI scripts get the designed routing logic. Paper forms are laid out for automated scanning (OCR) or manual data entry. For secondary and emerging sources, the receiving channels are configured. SFTP endpoints get the designed encryption and authentication, API connections get rate limiting and error handling, and data space connectors are set up according to their specifications. Secure transfer channels with end-to-end encryption are set up in line with SAF-P10. A respondent management system handles invitations, reminders, and tracking.

Activity	Key outputs	SAF process	Other reference
Build or configure survey instruments per mode (CAWI, CAPI, CATI, paper)	Production-ready instruments	PRD-3.1 Reuse or build collection instruments	1.1.3 Build statistics product
Configure data receipt channels (SFTP, API, data space connectors)	Configured channel endpoints	PRD-3.1 Reuse or build collection instruments; DSP-1 Join a data space	
Set up secure transfer channels with end-to-end encryption	Secure channel configuration	PRD-3.1 Reuse or build collection instruments	SAF-P10
Configure the respondent management system (invitations, reminders, tracking)	Respondent management system	PRD-3.1 Reuse or build collection instruments	3.4.2 Contact management

Configure Data Ingestion Workflows

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-2 Promote data through a quality gate.

This sub-process translates the designed validation rules and metadata capture requirements into automated workflows. Structural validation is implemented first: schema conformance, field types, and required fields. Alongside it come integrity checks (checksum verification, virus scanning, digital signature validation), completeness checks (record counts, field coverage thresholds), and service level checks (delivery timeliness, format conformance). Metadata capture at each ingestion point is automated, so that provenance, format, timestamps, and file hashes are extracted without manual work. The landing zone for raw data is set up with the designed access controls and storage policies. Event notifications are configured, so that VS-02 is alerted when raw data is registered. Exception handling workflows route rejected data to the right queue: provider notification, manual review, or conditional acceptance.

Activity	Key outputs	SAF process	Other reference
Implement validation rules as automated checks (structural, integrity, completeness, service level)	Automated validation pipeline	PRD-3.2 Reuse or build processing and analysis components	
Configure metadata capture at each ingestion point	Automated metadata extraction	PRD-3.4 Configure workflows	SAF-P09
Set up the raw data landing zone with access controls and storage policies	Configured repository	PRD-3.4 Configure workflows; DLM-4 Classify and protect data	
Configure event notifications for downstream consumers	Publication of the *Raw Data Registered* event	PRD-3.4 Configure workflows	SAF-P07
Configure exception handling workflows	Rejection routing and manual review queues	PRD-3.4 Configure workflows; DLM-2 Promote data through a quality gate	

Build Metadata Registration and Cataloguing

Refines: PRD-3.4 Configure workflows; DLM-3 Register and catalogue data; DLM-6 Provide data for reuse; OVA-2 Metadata Management.

This sub-process implements the metadata catalogue and the automated registration that make every dataset findable and traceable. Catalogue entries are built from the metadata model of *Design Data Model and Metadata Schema*. Metadata is propagated automatically into the catalogue from three sources: the collection instruments (variables, code lists, question wording), the transfer channels (source, method, timing), and the validation workflow (validation results, completeness scores). Metadata quality validation makes sure that every registered dataset has all required metadata fields. Search and discovery functions let downstream consumers find raw data and its metadata. As CSDA emphasises, the ultimate goal is active metadata: metadata that can generate collection instruments, validation rules, and processing specifications automatically.

Activity	Key outputs	SAF process	Other reference
Implement catalogue entries based on the designed metadata model	Populated metadata catalogue	DLM-3 Register and catalogue data; OVA-2 Metadata Management	

Activity	Key outputs	SAF process	Other reference
Configure automatic metadata propagation from instruments, channels, and validation	Automated metadata pipelines	PRD-3.4 Configure workflows	CSDA Metadata Management
Set up metadata quality validation	Metadata completeness checks	OVA-2 Metadata Management	SAF-P09
Implement metadata search and discovery for downstream consumers	Discovery interface	DLM-6 Provide data for reuse	CSDA Metadata Management

PLACEHOLDER: VS-01 implementation pipeline: the five design artefacts feeding the three build sub-processes, followed by technical testing, field test, and security and performance testing, resulting in a production-ready acquisition process and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-01 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when a collection period opens or when a provider's delivery arrives. The stages then run in order, but for continuous sources, such as daily web data, VS-01.3 to VS-01.5 repeat for every delivery. Execution applies the designs and the solution as they are; anything that needs a new rule or a new channel goes back to Design. Every delivery ends in one of three outcomes: registered, registered with quality flags, or returned to the provider.

Aspect	VS-01
Trigger	*Collection period opened*; *Delivery received*
Cadence	Per survey cycle; per delivery for administrative and web data
Performed by	ROL-13 Data collection manager, ROL-14 Data quality analyst, ROL-03 Data steward
Input	Source data (#0), active agreements, production-ready instruments
Output	Registered raw data version (#1) and its intake log
Control point	Gate of State #1 in VS-01.4
Exception path	Rejection to the provider via ROL-23 Relationship manager; structural change at the source back to Design

Management: provides quality reports

Management sub-process	Refines	Main outputs
Monitor Process Performance *(standard)*	OVA-4 Process Data Management; OVA-6 Data Supplier Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard; service level compliance report for all data providers (OVA-6, 3.4.3 Service agreement)
Assess Data Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Data quality report, quality scorecards, trend analysis
Manage Metadata Lifecycle	OVA-2 Metadata Management; DLM-8 Monitor data quality, lineage and use	Metadata quality report, catalogue maintenance log, lineage audit

Management sub-process	Refines	Main outputs
Audit and Compliance Reporting *(standard)*	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Audit trail from the external source to the raw data repository; records of processing activities (GDPR Article 30), reviews of data protection impact assessments for high-risk processing, reporting on data subject rights, and the ability to show which data the NSI holds and why; agreement compliance report confirming that all use stays within the agreements
Continuous Improvement *(standard)*	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, for example new sources that replace sources of lower quality, modes that reduce burden or raise response, automation of manual ingestion steps, and validation rules that produce fewer false alarms; trends watched: rejection rate, metadata completeness, processing time **[VC11]**

Assess Data Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality of the ingested data itself is evaluated against the criteria set in *Design Validation Rules and Quality Criteria*. Each dataset is assessed on several quality dimensions. Completeness covers record counts against expectations and field coverage. Timeliness compares the delivery date with the agreed deadline. Structural accuracy covers format conformance and valid value ranges. Consistency covers consistency between fields and with earlier deliveries. Conformance covers alignment with the designed schema and classification schemes. Quality indicators are produced per dataset, per source, and per production cycle. Quality trends are tracked to detect declining sources or modes with consistently lower quality. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Evaluate the quality dimensions per dataset: completeness, timeliness, accuracy, consistency, conformance	Data quality report (per cycle)	OVA-1 Quality Management	CSDA Data Quality
Compare the results with the designed quality criteria	Quality assessment against baseline	OVA-1 Quality Management	2.1.3 Statistical quality management
Produce quality indicators per dataset, per source, per cycle	Quality scorecards (per source, per mode)	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality
Identify quality trends across sources and modes	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management

Manage Metadata Lifecycle

Refines: OVA-2 Metadata Management; DLM-8 Monitor data quality, lineage and use.

The metadata that supports data acquisition must stay complete, accurate, and current throughout the data life cycle. This sub-process verifies metadata completeness: every registered dataset must have all required metadata fields, as a compliance check on SAF-P09. It assesses metadata quality: are provenance records accurate, are classifications applied correctly, and are agreement references current? The metadata catalogue is actively maintained. Outdated entries are retired, schemas are updated for new source formats, and metadata models are versioned. Lineage from source through ingestion to raw data is tracked. Reconciliation across systems makes sure that catalogue entries match the artefacts actually stored. As CSDA emphasises, metadata carries all the context needed to understand and reuse data; without it, data is effectively useless.

Activity	Key outputs	SAF process	Other reference
Verify metadata completeness for all registered datasets	Metadata quality report	OVA-2 Metadata Management	SAF-P09
Assess metadata quality: accuracy, currency, correct classification	Metadata quality report	OVA-2 Metadata Management	CSDA Metadata Management
Maintain the metadata catalogue: retire entries, update schemas, version models	Catalogue maintenance log	OVA-2 Metadata Management	CSDA Metadata Management
Track metadata lineage from source through ingestion to raw data	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Metadata Management
Reconcile metadata across systems	Catalogue-artefact reconciliation	OVA-2 Metadata Management	SAF-P09

PLACEHOLDER: VS-01 management feedback loop: data and paradata from Execution flowing through the five management sub-processes, with dashboards to the process owner, quality reports to the quality manager, compliance reports to VS-07, and improvement recommendations back to Design

Metadata per process type

Metadata is produced in every process type, not only in Design. The table shows the data metadata each process type of VS-01 contributes; the paradata are listed in Standard activities in every value stream. In Execution, VS-01 also records the collection mode and instrument version as paradata.

Process type	Data metadata produced
Design	Data models, variable definitions, classification schemes, expected formats, quality thresholds
Implementation	Schema configurations, parameters of validation rules, catalogue templates, mappings from instruments to metadata
Execution	Provenance, source, timestamp, file hash, volume, quality indicator values, and agreement reference per dataset
Management	Quality scores, lineage assessments, completeness reports, life cycle status, trend analyses

PLACEHOLDER: VS-01 business object lifecycle: design artefacts govern implementation assets, which produce the process that turns Source Data (#0) into Raw Data (#1), generating provenance, quality, and audit records that management assesses and feeds back to design

5. Actors and roles

VS-01 involves more external actors than any other value stream, because it is where the NSI meets its data providers. The table shows the actors and the reference roles involved. Specialised role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the system configurator in the original VS-01 description is ROL-11 Platform engineer. The long-term relationship with each provider is organised in the collaboration COL-01 *Data provision partnership*.

Actor or role	ID	Process type	Responsibility in VS-01
Data provider	ACT-03	External	Supplies data under a data sharing agreement: a government agency, a business, or an individual
Respondent	ACT-04	External	Provides data through survey instruments: web (CAWI), face-to-face (CAPI), telephone (CATI), or paper
Supervisory and legal authority	ACT-08	External	Grants or oversees the legal basis for collection, for example under the statistics law or data protection law
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-01
Data owner	ROL-02	Management	Accountable for State #1; decides on access and retention
Data steward	ROL-03	Execution	Registers datasets in the catalogue, manages metadata entries, applies access controls, resolves gate exceptions
Data custodian	ROL-04	Execution	Operates storage, access control, and retention
Statistical designer	ROL-06	Design	Translates statistical needs into data requirements and process specifications
Methodologist	ROL-07	Design	Designs collection methodology, sampling strategies, and questionnaire content
Metadata and data architect	ROL-08	Design	Designs data models, metadata schemas, and classification mappings
Data engineer	ROL-10	Implementation	Builds and configures ingestion workflows, collection instruments, and data channels

Actor or role	ID	Process type	Responsibility in VS-01
Platform engineer	ROL-11	Implementation	Configures platforms, SFTP endpoints, API connectors, and access controls
Test manager	ROL-12	Implementation	Plans and carries out testing of the acquisition process: technical tests, field tests, integration tests
Data collection manager	ROL-13	Execution	Oversees collection campaigns, coordinates collection teams, and monitors response
Data quality analyst	ROL-14	Execution	Performs validation checks, assesses data quality at intake, and logs exceptions
Quality manager	ROL-18	Management	Monitors key performance indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures legal and regulatory compliance and produces audit reports
Relationship manager	ROL-23	Execution	Maintains provider relationships and agreements; handles rejected deliveries with the provider

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-01 relies on, and what role they play. NSIs can use it to map the value stream to their own organisation.

Capability group	Capabilities	Role in VS-01
1.1 Statistical design and build	1.1.1 Needs inventory, 1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the data requirements and the collection methodology, and build the acquisition process
1.2 Data collection	1.2.1 Primary data collection, 1.2.2 Secondary data collection	Carry out the collection: surveys, intake of administrative data, extraction through APIs
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage data and metadata during ingestion, assess quality, maintain registers
3.1 Business operations	3.1.2 Purchasing and contract management	Manage contracts with data providers that are suppliers
3.2 Computerisation and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain collection instruments and ingestion systems

Capability group	Capabilities	Role in VS-01
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.3 Security incident management, 3.3.4 Logical identity and access control	Build privacy and security into all acquisition processes
3.4 Customer and chain interaction	3.4.2 Contact management, 3.4.3 Service agreement	Maintain relationships with respondents and providers, and manage data provision agreements
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, plan improvements, report on compliance

Stages, processes, and functions

The table shows which SAF business processes realise each stage of VS-01 and which business functions they mainly use.

Stage	Realised by processes	Main functions
VS-01.1 Identify Data Needs and Sources	PRD-1.1 Identify needs; PRD-1.5 Check data availability and suitability; PRD-2.3 Design collection	FUN-3.1 Design and build statistical products; FUN-6.1 Manage data supplier relationships
VS-01.2 Establish Data Agreements and Access Rights	OVA-6 Data Supplier Management; DSP-1 Join a data space; DSP-3 Agree data use conditions	FUN-6.2 Manage data provision agreements; FUN-9.1 Manage data space participation; FUN-10.5 Manage legal affairs
VS-01.3 Collect or Receive Data	PRD-4.1 Create frame and select sample; PRD-4.2 Set up collection; PRD-4.3 Run collection; DSP-4 Exchange data under usage control	FUN-3.2 Collect and acquire data; FUN-9.2 Operate data space connectivity
VS-01.4 Validate and Log Data Intake	PRD-4.4 Finalise collection; DLM-2 Promote data through a quality gate	FUN-3.2 Collect and acquire data; FUN-7.3 Manage data lifecycle and storage
VS-01.5 Store and Register Raw Data Artefacts	DLM-3 Register and catalogue data; DLM-4 Classify and protect data; DLM-5 Store and version data	FUN-7.2 Manage metadata; FUN-7.3 Manage data lifecycle and storage; FUN-7.5 Protect data

PLACEHOLDER: ArchiMate view VS-01 (b) stages realised by processes and functions

Framework mappings

This reference table maps each process type of VS-01 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-1.1 to PRD-1.3, PRD-1.5, PRD-1.6, PRD-2.2 to PRD-2.5, DLM-1, DLM-2, DLM-4, DLM-9, OVA-1, OVA-2, OVA-4, OVA-6	Phase 1 Specify Needs (1.1 to 1.3, 1.5, 1.6), Phase 2 Design (2.2 to 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management	1.1.1 Needs inventory, 1.1.2 Design statistics product, 2.1.2 Metadata management
Implementation	PRD-3.1, PRD-3.2, PRD-3.4 to PRD-3.7, DLM-2, DLM-3, DLM-4, DLM-6, OVA-2, DSP-1	Phase 3 Build (3.1, 3.2, 3.4 to 3.7)	Capability Management	Data Ingestion (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-1.1, PRD-1.5, PRD-4.1 to PRD-4.4, DLM-2 to DLM-5, OVA-2, OVA-4, OVA-6, DSP-1, DSP-3, DSP-4	Phase 4 Collect (4.1 to 4.4); 1.1 and 1.5 in their recurrent form	Production (execution)	Data Ingestion (execution)	1.2.1 Primary data collection, 1.2.2 Secondary data collection
Management	OVA-1, OVA-2, OVA-4, OVA-6, DLM-8, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-01 most directly.

Principle	Statement	Application in VS-01
SAF-P01	Actively participate in open-source communities to maintain collective control over statistical processes	Collection instruments, connectors, and ingestion tooling are reused across products first, in the order reuse, open source, buy, build. Where new tooling is needed, open-source solutions developed with other NSIs come first, and the NSI contributes back, so that it keeps collective control over critical capabilities instead of depending on vendors. **[VC3]**
SAF-P07	Implement loosely coupled processes and systems	Data acquisition is independent from downstream processing. Raw Data (State #1) is the decoupling interface: downstream consumers use it without knowing how it was collected.
SAF-P08	Data is shared property	Raw data, once registered, is available to all authorised consumers. Collection is not duplicated across statistical products when the same source data can serve several purposes. This also reduces response burden.
SAF-P09	No data without metadata and classification	At ingestion, every dataset is enriched with provenance (where it came from and how), source, method, timestamp, quality indicators, and a classification for availability, integrity, and confidentiality. This is not a precondition that blocks ingestion, but an obligation the NSI fulfils during acquisition.
SAF-P10	Security by design	Security is built into the acquisition process from the start: encrypted transfer channels, virus scanning, checksum verification, and access controls.
SAF-P11	Privacy by design	Privacy protection is designed into the acquisition process from the start, not added after data has entered: the moment of pseudonymisation, purpose limitation, the legal basis for processing, and data minimisation.

i Principles that apply across the process types

Several principles apply to all four process types. SAF-P09 applies everywhere: Design defines the target metadata, Execution captures it at ingestion, and Management assesses it and closes remaining gaps. SAF-P10 applies to *Design Access Control and Privacy Framework*, to *Build Collection*

Instruments and Channels, and to the stages VS-01.3 to VS-01.5. SAF-P11 starts in *Design Access Control and Privacy Framework* and carries through all process types. SAF-P07 governs the handover of State #1 to VS-02.

8. Interfaces, services, and events

Boundary interfaces

VS-01 has two boundary interfaces: intake from outside the NSI, and handover to the next value stream.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
Provider or respondent → VS-01 (intake)	Delivery or survey response (#0)	Source identifier and organisation, transfer method and channel, receipt timestamp, file hash, volume, agreement reference, collection mode and instrument version (surveys)	*Delivery received* or response submitted	The provider is responsible up to receipt; VS-01 takes custody and the obligation to capture metadata (SAF-P09) at the point of receipt
VS-01 → VS-02 (or VS-03) (handover)	Registered raw data version (#1)	Minimum metadata set of State #1: provenance, source, collection method, receipt timestamp, file hash, agreement reference, quality assessment, classification	*Raw Data Registered*	VS-01 keeps custody of the raw artefact; the consumer takes responsibility for further processing on receipt of the event

Business services

Service	Offered to	Realised in	Interfaces
Data collection service	Respondents, data providers	VS-01.3 Collect or Receive Data	Web portals (CAWI), mobile applications (CAPI), telephone systems (CATI), paper, SFTP endpoints, REST APIs, data space connectors
Data agreement service	Data providers, legal authorities	VS-01.2 Establish Data Agreements and Access Rights	Agreement management, relationship manager
Raw data provisioning service	VS-02 Data Standardization and Integration, VS-03 Data Editing and Enrichment, other domains	VS-01.5 Store and Register Raw Data Artefacts	Event notification (*Raw Data Registered*) and a catalogue query interface to find and access raw datasets

Service	Offered to	Realised in	Interfaces
Ingestion monitoring service	Process owner, quality manager	Management sub-processes *Monitor Process Performance* and *Assess Data Quality*	Dashboards and quality reports
Metadata catalogue service	All data consumers in the NSI	Management sub-process *Manage Metadata Lifecycle*	Search and browse interface for metadata, with metadata exchange compliant with SDMX
Design and methodology service	Methodologists, statistical designers, data engineers	All Design sub-processes of VS-01	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artefacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Collection period opened	Start of VS-01.3 for a survey cycle (PRD-4.3 Run collection)
Agreement expired or changed	A review in VS-01.2 before the next delivery
Delivery received	Start of VS-01.4 for that delivery
Data Intake Validated / Data Intake Rejected	Outcome of VS-01.4; rejection starts provider follow-up
Raw Data Registered	Completion of VS-01.5; triggers the consuming value stream
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-01 event choreography between data provider, NSI teams, and process management across Design, Implementation, Execution (agreement, primary or secondary collection, validation passed or failed), and Management (key performance indicators, quality, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-01	Receives from VS-01
VS-07 Governance and Compliance	Legal register, privacy and security policy	Audit trails, compliance evidence
VS-08 Innovation and Methodological Development	New sources and collection methods	Improvement needs
VS-09 Infrastructure and Platform Enablement	Channels, storage, catalogue, event interfaces	Capacity and channel requirements
VS-10 Customer and Stakeholder Engagement	Provider relationships and burden insight	Provider feedback and issues

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
Response rate	Share of sampled units that provided data (surveys)	Per survey, per mode, per cycle
Delivery timeliness	Share of deliveries received within the agreed window	Per source, per cycle
Rejection rate	Share of deliveries that failed the gate of State #1	Per source, per cycle
Processing time	Time from receipt to catalogue registration	Per delivery, per cycle
Metadata completeness	Share of required metadata fields populated at registration	Per delivery
Exception rate	Frequency of exception handling per check rule	Per rule, per cycle

10. Examples

Survey. The Labour Force Survey uses a mixed-mode approach and a 2-2-2 rotation panel for longitudinal tracking. Sampled persons first receive a web invitation by email; non-respondents are called after two weeks, and hard-to-reach groups are visited by an interviewer. Adaptive questioning keeps response burden low. Each mode records its own paradata, such as completion time or interviewer identifier. Responses are registered as one #1 version per collection batch, and the data collection manager follows the response rate per mode during the cycle.

Multi-mode census. The 2026 census collects 70% of households by web (CAWI), 25% by interviewer (CAPI) for non-respondents and hard-to-reach areas, and 5% on paper for elderly and institutionalised populations. Each mode records its own metadata: browser type and completion time for CAWI; interviewer identifier, GPS coordinates, and interview duration for CAPI; scanning batch identifier and OCR confidence scores for paper.

Household budget survey. The implementation team builds a CAWI portal with a diary for daily spending, configures it in 12 languages, and implements the designed sampling filter. It also sets up an SFTP channel as a fallback for large retailers, which deliver scanner data as supplementary price information.

Administrative register. The agreement with the tax authority requires delivery within 45 days of quarter end and at least 98% of the expected records. Before the Q3 delivery window opens, the team checks that the SFTP credentials are valid and records the active agreement reference, to be linked to the incoming file. The file arrives by secure file transfer on day 38, with 2.3 million records, 99.2% of the expected volume. The SHA-256 checksum passes, all 47 fields conform to the schema, and the null rate on mandatory fields is 0.3%, within the 1% tolerance. Fourteen unknown business identifiers are not an intake issue; they pass through with a flag. The result is *Data Intake Validated*, and every check is logged with its timestamp, rule version, and operator identifier. The file is registered as `RD-TAX-2026Q3-001`, and the *Raw Data Registered* event notifies VS-02.

Register-based census. For its next census, the NSI chooses a register-first design. The population register is the primary source. A sample survey adds variables that registers do not hold, such as housing conditions and commuting patterns, and municipal records are used for validation. Design produces the mapping from register variables to census variables, the sample design of the supplementary survey, and the quality criteria for accepting register data. At the start of each reference year, the census team confirms that all 393 municipal registers are available, verifies contact persons and API endpoints, and checks that the survey frame has been updated with the latest extract of the population register.

Web data. Every night, 12,500 product pages of 25 retailers are scraped; on a typical day 95% succeed. Each scrape records the scraper version and a fingerprint of the page structure. When one retailer redesigns its website, the fingerprints change and the success rate drops. Management detects the trend and sends an improvement action to Design, where the scraper is rebuilt.

Toolkit: SAF-TK-V01 VS-01 maturity checklist and scenarios